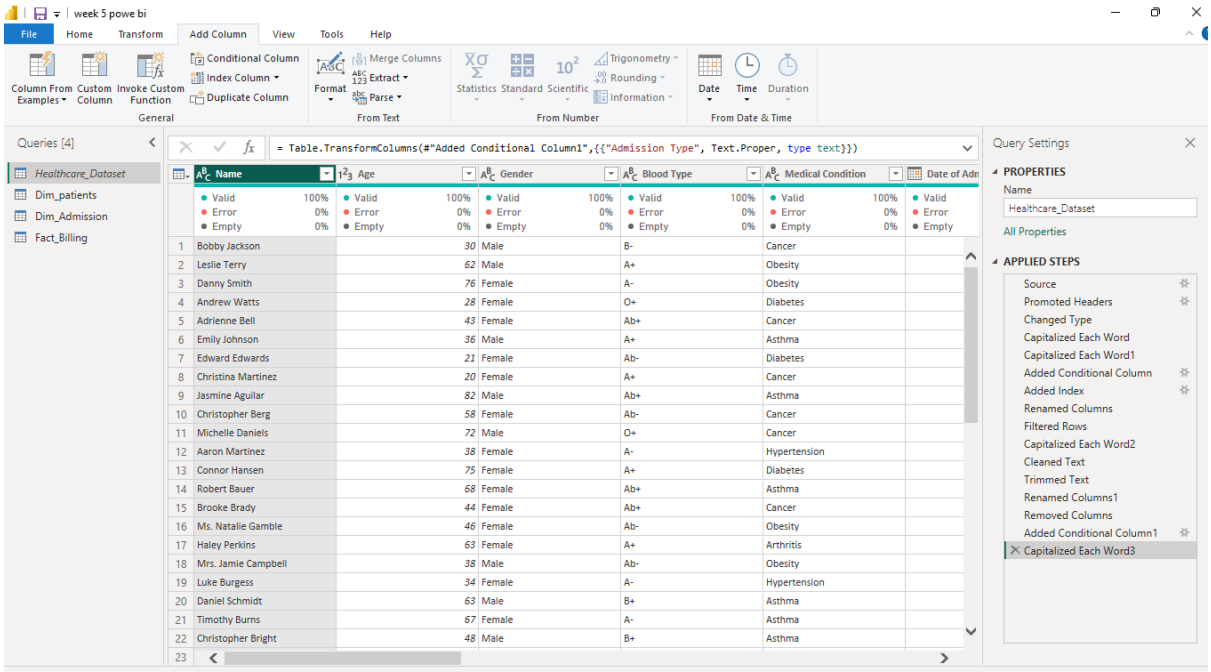
WEEK 5 – DATA CLEANING AND DATA MODELING USING POWER BI



1. Objective

The main objective of this week is to clean and transform the healthcare dataset and create a structured data model using Power BI. The dataset was prepared by checking data quality, creating useful categories, and dividing the data into dimension and fact tables.

2. Tools Used

- **Microsoft Power BI**
- **Power Query Editor**
- **Healthcare Dataset**

3. Data Cleaning

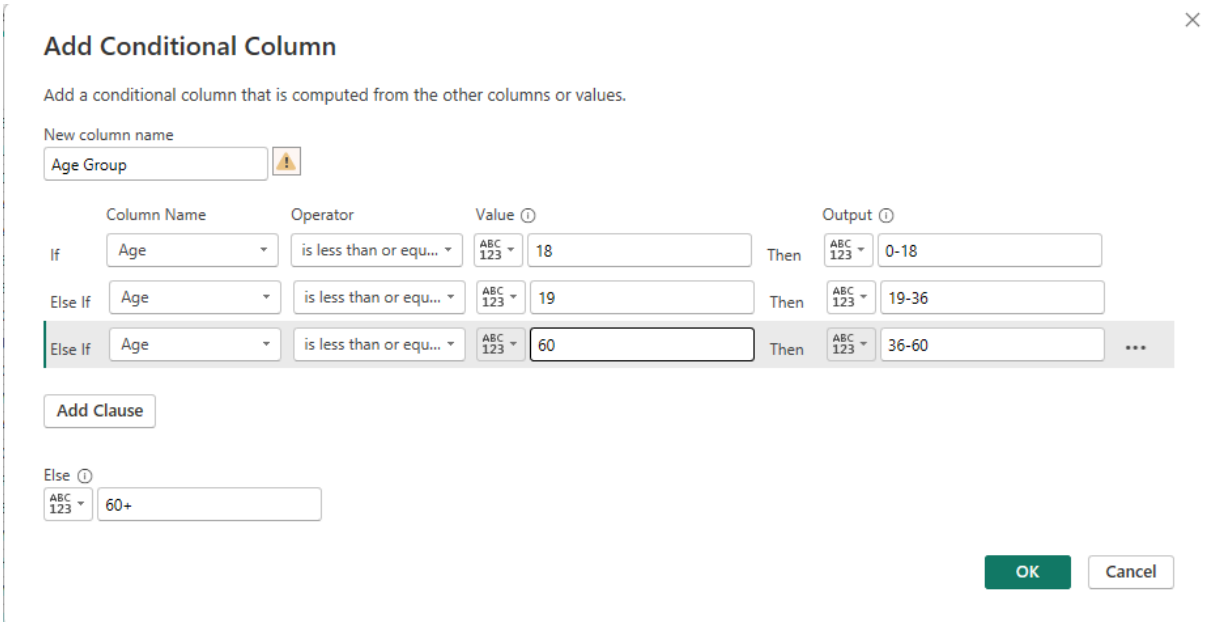
The healthcare dataset was loaded into Power BI and opened in Power Query Editor.

The following data cleaning operations were performed:

- Checked the dataset for null values.
- No null values were found in the dataset.
- Standardized the **Admission Type** values.
- Created a new **Age Group** column based on patient age.

The age groups created were:

Age Range	Age Group
0–18	0-18
19–35	19-35
36–60	36-60
Above 60	60+



4. Creating Dimension and Fact Tables

To organize the healthcare data efficiently, three tables were created using Power Query Reference:

Dim_Patients

This table contains patient-related information such as:

- Name
- Age
- Age Group
- Blood Type
- Gender
- Medical Condition
- Patient ID

A unique **Patient ID** was created using an Index Column.

Dim_Admission

This table contains admission-related information such as:

- Admission Type
- Date of Admission
- Discharge Date
- Doctor
- Hospital
- Insurance Provider
- Room Number
- Admission ID

A unique **Admission ID** was created using an Index Column.

Fact_Billing

This table contains billing and medical transaction-related information such as:

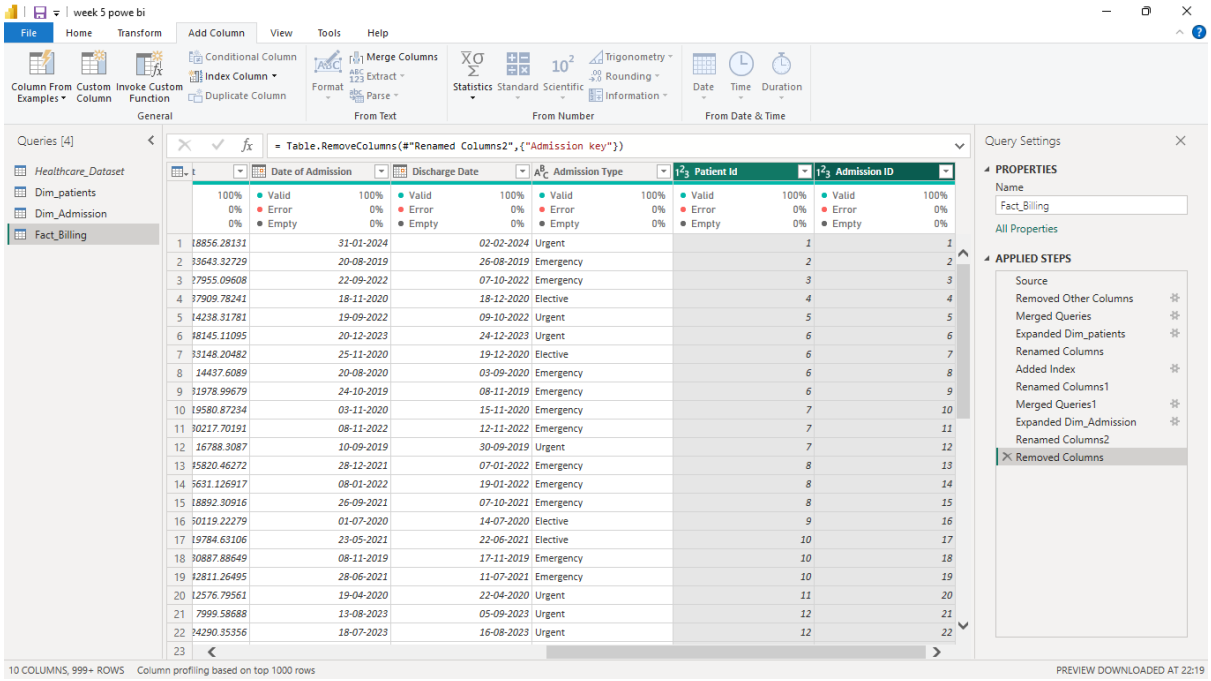
- Medication
- Test Results
- Billing Amount
- Date of Admission
- Discharge Date
- Admission Type
- Patient ID
- Admission ID
- Name
- Medical Condition

5. Merge Queries

Merge Queries was used to connect the fact table with the dimension tables.

- Dim_Patients was merged with Fact_Billing using **Name** to bring the **Patient ID**.
- Dim_Admission was merged with Fact_Billing using the admission key to bring the **Admission ID**.

This helped create proper keys in the Fact_Billing table.



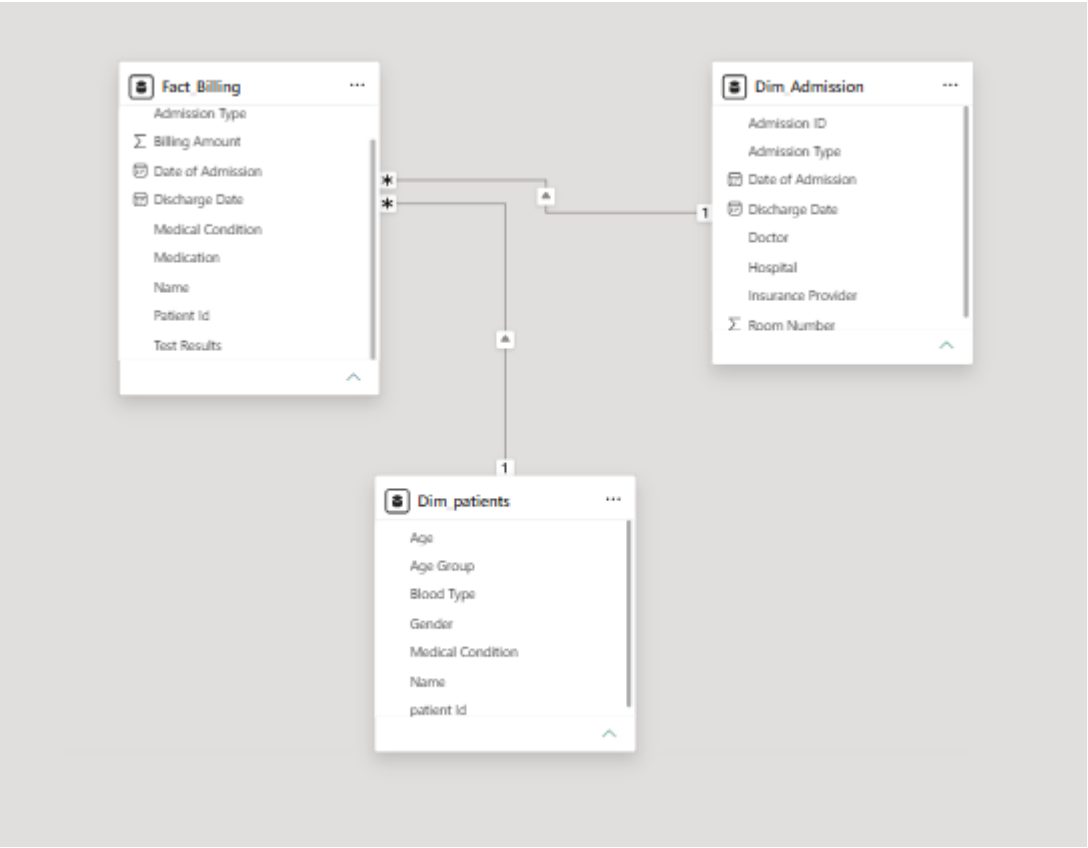
6. Data Model and Relationships

The final data model was created in **Model View**.

The following relationships were established:

- **Dim_Patients (1) → Fact_Billing (*)**
- **Dim_Admission (1) → Fact_Billing (*)**

The direct relationship between Dim_Patients and Dim_Admission was removed to avoid unnecessary relationships.



7. Result

The healthcare dataset was successfully cleaned, transformed, and organized into dimension and fact tables. The final model contains proper relationships between patient, admission, and billing information.

8. Conclusion

In Week 5, the healthcare dataset was prepared for analysis using Power BI. Data cleaning, age grouping, table creation, merging, and relationship modeling were successfully completed. The final structured model provides a suitable foundation for developing interactive dashboards and visualizations in the next stage.